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Functions of actin-interacting protein 1 (AIP1)/WD repeat protein 1 (WDR1) in actin filament dynamics and cytoskeletal regulation

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Abstract

Actin-depolymerizing factor (ADF)/cofilin and actin-interacting protein 1 (AIP1), also known as WD-repeat protein 1 (WDR1), are conserved among eukaryotes and play critical roles in dynamic reorganization of the actin cytoskeleton. AIP1 preferentially promotes disassembly of ADF/cofilin-decorated actin filaments but exhibits minimal effects on bare actin filaments. Therefore, AIP1 has been often considered to be an ancillary co-factor of ADF/cofilin that merely boosts ADF/cofilin activity level. However, genetic and cell biological studies show that AIP1 deficiency often causes lethality or severe abnormalities in multiple tissues and organs including muscle, epithelia, and blood, suggesting that AIP1 is a major regulator of many biological processes that depend on actin dynamics. This review summarizes recent progress in studies on the biochemical mechanism of actin filament severing by AIP1 and *in vivo* functions of AIP1 in model organisms and human diseases.

Keywords

ADF/cofilin; actin dynamics; cytoskeleton; severing

1. Introduction

Dynamic reorganization of the actin cytoskeleton is often necessary for many biological processes [1]. To achieve cytoskeletal reorganization, both assembly and disassembly of actin filaments need to be spatially and temporally controlled. Since polymerized actin is relatively stable, specific sets of actin-regulatory proteins are required to promote actin filament disassembly [2,3]. Actin filament disassembly is important not only for removal of old actin filaments but also to replenish the actin monomer pool that is used for a new round of actin assembly. Therefore, coordinated regulation of actin assembly and disassembly allows persistent cytoskeletal reorganization.

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Conflict of interest

The author declares no conflict of interest.

Actin depolymerizing factor (ADF)/cofilin promotes actin disassembly by severing actin filaments and dissociating actin monomers from filaments, and its significant roles in actin cytoskeletal remodeling have been demonstrated in a variety of cell types across eukaryotes [4,5,6]. Although ADF/cofilin is often accumulated at subcellular regions where the actin cytoskeleton is dynamic, ADF/cofilin alone, in particular at high concentrations, is not very efficient to drive rapid actin filament dynamics [7,8]. Therefore, additional factors need to cooperate with ADF/cofilin when rapid actin turnover is required. ADF/cofilin preferentially binds to ADP-bound actin and inhibits the exchange of ADP with ATP on actin monomers [9,10,11]. This process is antagonized by profilin or cyclase-associated protein to promote nucleotide exchange and produce ATP-actin monomers, which is highly competent for polymerization [12]. To sever actin filaments, clustered binding of ADF/cofilin to actin filaments is required to accumulate sufficient structural changes at a boundary between ADF/cofilin-bound and bare regions, where fragmentation preferentially occurs [8]. Therefore, ADF/cofilin does not sever filaments efficiently at high concentrations, at which ADF/cofilin saturates actin filaments without generating a structural discontinuity [13]. Actin-interacting protein 1 (AIP1), which is also known as WD-repeat protein 1 (WDR1), strongly enhances actin filament disassembly even when actin filaments are saturated by ADF/cofilin [14]. Profilin, cyclase-associated protein, and AIP1 are also present widely among eukaryotes and participate in a variety of actin-dependent events. This review focuses on AIP1 and summarizes recent progress in characterization of biochemical and physiological functions of AIP1, which have not been discussed in previous review articles [2,3,14].

2. Nomenclature of AIP1/WDR1 across eukaryotes

The original actin-interacting protein 1 (Aip1p) was discovered and cloned by Amberg and colleagues from budding yeast as one of proteins that interact with actin in a yeast two-hybrid screen [15]. Since then, a number of orthologs have been identified across eukaryotes and given a variety of gene/protein names, which are summarized in Table 1. Therefore, I propose to use “actin-interacting protein 1 (AIP1)” as a common name for these orthologous proteins along with species-specific gene or protein names to avoid confusion and maintain traceable history of publications through literature search. Vertebrate AIP1 was reported as WD-repeat (also known as WD40-repeat) protein 1 (WDR1) in chicken, mouse, and human [16], which was designated by the presence of multiple WD repeats in the primary sequences (see below). WDR1 has become a common name for vertebrate AIP1s in most of databases. In *Drosophila melanogaster* and *Caenorhabditis elegans*, genetic loci for AIP1 genes were identified by mutant phenotypes before the genes were molecularly characterized. The *Drosophila flare* gene encodes AIP1 [17], because *flare* mutations cause abnormal flare-shaped wing hairs. The *C. elegans unc-78* gene encodes AIP1 [18], since mutant worms exhibit uncoordinated movement with severe muscle defects [19]. *C. elegans* has a second AIP1 gene, *aip1-1* (*AIP1-like protein-1*), which has partially redundant function with *unc-78* [20].

3. Sequence and structure of AIP1

Molecular architecture of AIP1 proteins is highly conserved across eukaryotes. AIP1s are ~600 amino acids long (65–67 kDa) containing multiple WD repeats. WD repeats are found in many proteins with diverse functions, and, typically, seven WD repeats are arranged into a single β -propeller domain with each WD repeat forming a “blade” consisting of a four-stranded antiparallel β -sheet [21]. Crystal structures of budding yeast AIP1 [22] and *C. elegans* UNC-78 [23] revealed two seven-bladed β -propeller domains in a single AIP1 molecule (Fig. 1A), indicating that each AIP1 molecule has 14 WD repeats, some of which could not be identified from consensus sequence analysis. The two β -propeller domains are attached rigidly by multiple hydrogen bonds in a twisted manner, which generates concave and convex surfaces (Fig. 1B). The hydrogen bonds between backbones and side chains of the two β -propeller domains do not appear to allow flexibility in the angle between the two propellers. However, if a post-translational modification or binding partner could alter the angle between the propellers, it might significantly alter the function of AIP1.

4. Mechanism of actin filament disassembly by ADF/cofilin and AIP1

AIP1 preferentially interacts with ADF/cofilin-decorated actin filaments and enhances filament disassembly [24,25,26,27]. Mutational studies show that most of functional residues required for actin disassembly are located on the concave surface of the molecule (Fig. 1B) [23,28,29,30,31]. These residues are also essential for the function of AIP1 to regulate actin cytoskeleton *in vivo* [28,30,31,32,33]. A molecular docking simulation shows that ADF/cofilin fits on the concave surface of AIP1 (Fig. 1B), suggesting that ADF/cofilin mediates binding of AIP1 to actin filaments [30]. Direct observations of actin filaments with light microscopy demonstrate that AIP1 promotes severing of ADF/cofilin-decorated actin filaments [34,35,36,37,38,39]. Single molecule analysis using labeled AIP1 has confirmed that indeed AIP1 has little interaction with actin filaments in the absence of ADF/cofilin, and that it specifically targets ADF/cofilin-decorated regions of filaments, where it induces severing events rapidly upon binding [36]. A kinetic analysis shows that a single AIP1 molecule severs an ADF/cofilin-decorated actin filament within one second [K. Hayakawa, C. Sekiguchi, M. Sokabe, S. Ono, and H. Tatsumi, submitted], suggesting that AIP1 is a highly efficient severing protein. In contrast, ADF/cofilin alone only moderately severs actin filaments [40,41]. When ADF/cofilin binds to an actin filament at a high density, it remains bound to the filament for several minutes without severing [42]. Filament severing by ADF/cofilin alone occurs optimally in the presence of low concentrations of ADF/cofilin [13,43,44]. Under these conditions, cooperative binding of ADF/cofilin to actin generates ADF/cofilin-bound and bare clusters of actin subunits within a filament [45,46], thereby increasing severing at or near the boundaries between ADF/cofilin-bound and bare regions due to mechanical and structural heterogeneity [42,44,47,48].

Two hypotheses for the mechanism of actin filament severing by AIP1 have been proposed. One hypothesis is that AIP1 competitively displaces ADF/cofilin from actin to generate ADF/cofilin-decorated and bare boundaries, which favors severing (Fig. 2A) [49]. Myosin and phalloidin, which are known competitors of ADF/cofilin, can reduce the occupancy of ADF/cofilin on actin filaments and promote severing [49]. However, conflicting

experimental data on the competition between AIP1 and ADF/cofilin have been reported [35,37], and further investigation is needed to test this hypothesis. Another hypothesis is that AIP1 modulates a local actin-ADF/cofilin interface to activate severing (Fig. 2B) [50]. ADF/cofilin has two actin-binding interfaces, and one of them, which includes the F-loop and the C-terminal helix, is specifically required for its binding to F-actin [51,52,53]. Aggeli et al. have recently found that charge-reversal mutations of ADF/cofilin at the F-actin-binding site enhances its actin severing activity [50]. Therefore, if AIP1 induces a conformational change in the F-actin-binding site of ADF/cofilin, it can activate filament severing. However, to date, there is no direct evidence of such an AIP1-induced conformational change on ADF/cofilin. These two mechanisms are not mutually exclusive. For example, AIP1 may induce conformational change of ADF/cofilin to enhance its dissociation. High-resolution structure of the AIP1-ADF/cofilin-actin complex should provide insight in the mechanism of filament severing. Interestingly, three human ADF/cofilin isoforms differently mediate actin disassembly in the presence of AIP1 and coronin: cofilin-2 and ADF are more effective than cofilin-1 to enhance disassembly [54]. Two *C. elegans* ADF/cofilin isoforms, UNC-60A and UNC-60B, with different affinity with F-actin [55,56] have a significant difference in the orientation of the F-loops [57] and exhibit different effects in mediating AIP1-dependent actin disassembly [20,58]. Therefore, further analysis of ADF/cofilin isoforms may also provide a clue to understanding the mechanism of actin filament severing by AIP1.

5. Modulatory mechanisms of AIP1-dependent actin filament disassembly

Since AIP1 depends on ADF/cofilin to enhance actin filament disassembly, a variety of signaling pathways that regulates ADF/cofilin activity by phosphorylation and dephosphorylation [59,60] should also indirectly affect AIP1-dependent actin disassembly. Besides these indirect mechanisms, the AIP1 activity is directly modulated by pH: AIP1 is inactive under slightly acidic conditions (~pH 6.5) and becomes progressively active under neutral to basic conditions (> pH 7.5) [39]. ADF/cofilin also exhibits pH sensitivity in a similar pH range [10,11,61,62]. The pH sensitivity of AIP1 and ADF/cofilin is consistent with their roles in promoting pH-sensitive chemotaxis in *Dictyostelium* cells [33], in which an elevation of intracellular pH is required to enhance cell migration. In mammalian cells, elevated intracellular pH also enhances actin dynamics and cell migration [63]. Therefore, whether AIP1 is a part of a pH-sensing mechanism in cell migration in other cell types is a question for future investigation.

Actin filament disassembly by the combination of ADF/cofilin and AIP1 is significantly enhanced by the further presence of coronin [36,64,65] or caspase-11 [66] by distinct mechanisms. Coronin binds to actin filaments and enhances ADF/cofilin binding to the filaments [36,64,67,68], which are then rapidly disassembled by AIP1 [36]. In contrast, caspase-11 directly binds to AIP1 and actin to accumulate AIP1 on ADF/cofilin-decorated actin filaments [66]. In particular, the coronin-ADF/cofilin-AIP1 trio is highly efficient and capable of rapidly disassembling branched actin network such as *Listeria* comet tails [64] or individual actin filaments [36]. Coronin is widely present among eukaryotes (but absent in plants) [69], and genetic studies support cooperative roles of coronin and AIP1 in actin dynamics in yeast [70] and *Dictyostelium* [71]. However, there remain key discrepancies to resolve. For instance, Kueh et al. reported that the three proteins induce bursting

disassembly, which involves rapid disintegration along the filaments [65], whereas Jansen et al. reported strong severing of coronin-ADF/cofilin-decorated filaments by AIP1, which is followed by capping of the barbed ends of severed filaments [36].

Early studies showed that AIP1 caps barbed ends of ADF/cofilin-bound actin filaments, such that capping prevents reannealing of ADF/cofilin-severed filaments and enhances disassembly [72,73]. However, recent studies have consistently concluded that AIP1 does not cap barbed ends of ADF/cofilin-decorated actin filaments *in vitro* [34,35,36,37,74]. Jansen et al. demonstrated that coronin is required for AIP1 to cap barbed-ends of ADF/cofilin-decorated actin filaments [36]. Therefore, within cells, AIP1 can associate with barbed ends of actin filaments to limit filament growth [75]. Synergistic functions between AIP1 and capping protein in actin-dependent processes in yeast are consistent with the involvement of AIP1 in capping filaments *in vivo* [30,37,73,76,77].

Under stress conditions, ADF/cofilin and actin form intranuclear or cytoplasmic rod-like structures where AIP1 is also accumulated [26,78,79]. Within the ADF/cofilin-actin rods, both ADF/cofilin and actin filaments undergo very little dynamics [79,80]. Therefore, there should be a mechanism to stabilize the complex of ADF/cofilin, AIP1, and actin filaments without inducing disassembly. However, such a condition has not been reconstituted *in vitro*. This may be a part of stress response to minimize actin dynamics.

6. Biological functions of AIP1

Genetic studies show that AIP1 deficiency causes lethality in mice [81,82], insects [17], nematodes [20], and plants [83]. In addition, recessive point mutations in the human *WDR1* gene cause severe blood disorders [84,85]. These observations indicate that AIP1 plays an indispensable role in the regulation of actin cytoskeleton *in vivo*. In this section, *in vivo* functions of AIP1 in several representative tissues and organs in multicellular organisms are summarized and discussed.

6.1. Functions of AIP1 in epithelial cells

Epithelial tissues appear to be particularly sensitive to the function of AIP1. Avian *WDR1* was originally discovered as a mRNA that is up-regulated in the auditory epithelium upon acoustic damage and following regeneration [16]. The *WDR1* mRNA is also highly expressed in developing auditory epithelium, suggesting that it is important for morphogenesis of the inner ear [86]. *Drosophila* AIP1 mutants, *flare*, exhibit epidermal defects in hair morphology and planar cell polarity [17] and eye epithelial abnormalities in morphogenesis and junctional remodeling [87]. In *C. elegans*, AIP1 deficiency causes defects in assembly of contractile actin networks in the gonadal myoepithelial cells [88]. In agreement with these studies in invertebrate model organisms, a recent study in mouse epidermal development shows that *WDR1* plays critical roles together with ADF/cofilin in apicobasal and planar cell polarity by regulating cell shape dynamics and cortical tension [89]. In cultured mammalian intestinal epithelial cells, *WDR1* is accumulated at the epithelial junctions and plays essential roles in assembly of apical cell-cell junctions by promoting dynamics of perijunctional actin cytoskeleton, which is required for recruitment, internalization, and stabilization of junctional proteins [90]. Therefore, malfunction of AIP1/

WDR1 may compromise epithelial barriers in a variety of organs, which could underlie the cause of high susceptibility for infection and inflammation in human patients with mutations in the *WDR1* gene [84,85] (see below).

6.2. Functions of AIP1 in sarcomere assembly and maintenance in muscle cells

Essential role of AIP1 for sarcomeric actin assembly in striated muscle has been reported in *C. elegans* [18,20], *Drosophila* [91], and mouse [92], indicating that AIP1-dependent actin reorganization is a conserved requirement for myofibril assembly in animals. Although muscle sarcomeres are relatively stable contractile apparatuses, dynamic regulation of sarcomeric actin filaments is necessary for assembly and maintenance of myofibrils [93]. In *C. elegans*, mutations in the *unc-78* gene, which encodes AIP1, cause severe disorganization of sarcomeres in the body wall muscle with formation of actin aggregates [18,58]. Correlation of biochemical activities and *in vivo* phenotypes of AIP1 mutants indicate that the activity of AIP1 to disassemble ADF/cofilin-bound actin filaments is essential for proper sarcomere assembly [31]. *unc-78* acts synergistically with *unc-94*, which encodes an actin pointed-end capping protein tropomodulin, suggesting that both AIP1 and tropomodulin prevent excessive elongation of sarcomeric actin filaments [94]. The second *C. elegans* AIP1 gene, *aip1-1*, is partially redundant with *unc-78*: worms are viable with depletion of a single AIP1 isoform but arrested at late embryonic stages due to severe muscle defects with simultaneous depletion of the two AIP1 isoforms [20]. Consistent with these observations, in *Drosophila*, muscle-specific knockdown of AIP1 causes sarcomere disorganization in embryonic muscle [91]. In mice, cardiomyocyte-specific knockout of *WDR1* causes postnatal lethality with cardiac hypertrophy, contractility defects, and sarcomere disorganization with actin aggregate formation [92]. In human skeletal muscle, WDR1 protein levels are up-regulated during exercise after muscle disuse, suggesting that WDR1 is involved in adaptive sarcomere reorganization [95]. Further functional studies of AIP1 using complementary approaches in different organisms should reveal conserved significance of AIP1 in normal muscle physiology as well as muscle disease.

6.3. WDR1 mutations in blood disorders in vertebrates

Mutations in the mouse *WDR1* gene cause multiple blood disorders in mice and humans, indicating that AIP1 plays particularly important roles in the hematopoietic lineages. The first *WDR1* mutant mice, which were isolated from a screen for hematopoiesis-defective mutants, were called *redears* because they exhibit autoimmune lesions in their ears [81]. Enhanced release of interleukin-18 from monocytes triggers autoinflammation in *WDR1* mutants [96]. Neutrophils from *WDR1* mutant mice are defective in chemokine-induced actin reorganization and chemotaxis [81]. *WDR1* mutant mice also show reduced levels of platelets (thrombocytopenia) [81], as WDR1 has a role to promote shedding of platelets from megakaryocytes [97]. Platelets from *WDR1* mutant mice are defective in aggregation and stimulation-dependent actin remodeling [98]. Recent findings of mutations in the human *WDR1* gene confirmed that many of these mouse phenotypes are present in human patients [84,85]. Kuhns et al. identified 5 missense mutations in *WDR1* from 3 independent families and showed that individuals with homozygous or transheterozygous mutations exhibit recurrent infections, stomatitis, reduced neutrophils with impaired migratory activity [84]. This study suggests that *WDR1* malfunction may underlie the cause

of “lazy-leukocyte syndrome” with similar symptoms [99]. Standing et al. identified another missense mutation in *WDR1* from two siblings with autoinflammatory diseases, immunodeficiency, and thrombocytopenia but with relatively mild alterations in neutrophil functions [85]. The reported *WDR1* mutations in humans are recessive, causing childhood death in severe cases [84,85]. However, biochemical basis of these *WDR1* missense mutations remains uncharacterized. Whether any of these symptoms are caused or enhanced by compromised epithelial barriers is unknown. These studies suggest that blood cells are highly susceptible to defects in AIP1, and that AIP1-mediated rapid actin disassembly is crucial for the function of multiple types of blood cells.

6.4. Implications of WDR1 in cancer progression

ADF/cofilin plays a significant role in promoting invasive proliferation and cell migration of cancer cells by accelerating actin filament turnover [59,100,101]. Not surprisingly, mammalian WDR1 is also implicated in progression of cancer. Up-regulation of the WDR1 protein or mRNA level has been reported in several types of cancer, including breast cancer [102,103,104,105], glioblastoma [106,107], ovarian carcinoma [108], and thyroid carcinoma [109]. In breast cancer cell lines, transcription of WDR mRNA is enhanced by STAT3 [102], while WDR1 expression can be suppressed by the miR-206 microRNA, which is often down-regulated in cancer [110]. However, regulatory mechanisms of WDR1 protein and mRNA in other cancer types remain largely unknown. Studies in model organisms and cultured cells demonstrate that AIP1 promotes cell migration by enhancing lamellipodial actin dynamics [111,112,113,114], and that AIP1 knockdown or knockout impairs cytokinesis by causing malformation of the contractile ring where actin filaments accumulate excessively [111] and reduced myosin accumulation to the contractile ring [37]. AIP1 also promotes cell rounding during mitosis [115]. Therefore, positive regulation of cell migration and mitotic cell division by mammalian WDR1 may enhance cancer progression. If this is the case, WDR1 can be an attractive therapeutic target for cancer treatment. In addition, WDR1 is reported to enhance nuclear translocation of MRTF-A (also known as MKL1 or MAL), a co-factor of serum response factor, thereby altering gene expression in breast cancer cells [110]. However, in yeast, AIP1 deficiency is reported to decrease actin monomers [116]. Therefore, WDR1 is expected to increase actin monomers, which is known to inhibit nuclear translocation of MRTF-A by direct binding [117], suggesting that WDR1 may influence the nuclear translocation of MRTF-A by an unknown mechanism.

6.5. Function of AIP1 in plant development

AIP1 is also critical for plant development. *Arabidopsis thaliana* has two AIP1 genes, *AtAIP1-1* and *AtAIP1-2*, which are expressed in reproductive and all tissues, respectively [118]. Knockdown of both AIP1 isoforms by RNA interference arrests plant development and causes lethality [83]. Overexpression of *AtAIP1-1* causes formation of abnormal short and thick root hairs [119], while a mutation of *AtAIP1-2*, together with a mutation of an actin isoform ACTIN7, disrupts planar cell polarity of root hair positioning [120]. The rice *Oryza sativa* has only one AIP1 gene, and either overexpression or RNA interference of AIP1 decreases the size of adult plants [38]. In lily pollen tubes, AIP1 and ADF/cofilin localize to the growing tips and implicated in their elongation [121]. The moss *Physcomitrella patens* also has a single AIP1 gene, and AIP1 knockout causes significant

reduction of growth and formation of tip-growing cells [122]. At cellular levels, overexpression of AIP1 enhances actin turnover [38,119], whereas depletion of AIP1 stabilizes actin filaments and causes excessive bundling [38,83,122].

7. Perspectives

Recent progress in characterization of *in vivo* functions of AIP1 clearly indicates that AIP1 is not simply a supporting factor for ADF/cofilin. Rather, AIP1 and ADF/cofilin are functional partners. These two proteins collaborate to produce unique activities that neither one alone, nor any other actin-regulatory proteins, is capable of, including rapid severing and disassembly of filaments, and barbed-end capping in the presence of coronin. Lethality and severe phenotypes due to AIP1 deficiency show that AIP1 is essential for viability in a variety of organisms. Strong phenotypes in multiple tissues and organs in *WDR1* mutant mice prompt further examination of symptoms in human patients with *WDR1* mutations. While AIP1 appears to be largely redundant with capping protein in yeast, actin cytoskeleton is altered in different manners in each mutant, indicating their distinct roles. Although most of genetic observations are consistent with the function of AIP1 to disassemble ADF/cofilin-bound actin filaments, whether AIP1 has other ADF/cofilin-independent functions remains unknown. A recent study shows that mouse *WDR1* binds to LIM-kinase and enhances its activity to phosphorylate ADF/cofilin [82], which could result in stabilization, rather than disassembly, of actin filaments. Therefore, consequences of AIP1/*WDR1* up-regulation in cancer and other conditions need to be carefully evaluated in each case.

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Abbreviations

ADF	Actin-depolymerizing factor
AIP1	Actin-interacting protein 1
WDR1	WD-repeat protein 1

References

1. Pollard TD, Cooper JA. Actin, a central player in cell shape and movement. *Science*. 2009; 326:1208–1212. [PubMed: 19965462]
2. Ono S. Mechanism of depolymerization and severing of actin filaments and its significance in cytoskeletal dynamics. *Int Rev Cytol*. 2007; 258:1–82. [PubMed: 17338919]
3. Brieher W. Mechanisms of actin disassembly. *Mol Biol Cell*. 2013; 24:2299–2302. [PubMed: 23900650]
4. Bernstein BW, Bamberg JR. ADF/cofilin: a functional node in cell biology. *Trends Cell Biol*. 2010; 20:187–195. [PubMed: 20133134]
5. Kanellos G, Frame MC. Cellular functions of the ADF/cofilin family at a glance. *J Cell Sci*. 2016; 129:3211–3218. [PubMed: 27505888]

6. Mabuchi I. An actin-depolymerizing protein (depactin) from starfish oocytes: properties and interaction with actin. *J Cell Biol.* 1983; 97:1612–1621. [PubMed: 6226671]
7. De La Cruz EM, Gardel ML. Actin mechanics and fragmentation. *J Biol Chem.* 2015; 290:17137–17144. [PubMed: 25957404]
8. Elam WA, Kang H, De La Cruz EM. Biophysics of actin filament severing by cofilin. *FEBS Lett.* 2013; 587:1215–1219. [PubMed: 23395798]
9. Maciver SK, Weeds AG. Actophorin preferentially binds monomeric ADP-actin over ATP-bound actin: consequences for cell locomotion. *FEBS Lett.* 1994; 347:251–256. [PubMed: 8034013]
10. Hawkins M, Pope B, Maciver SK, Weeds AG. Human actin depolymerizing factor mediates a pH-sensitive destruction of actin filaments. *Biochemistry.* 1993; 32:9985–9993. [PubMed: 8399167]
11. Hayden SM, Miller PS, Brauweiler A, Bamburg JR. Analysis of the interactions of actin depolymerizing factor with G- and F-actin. *Biochemistry.* 1993; 32:9994–10004. [PubMed: 8399168]
12. Ono S. The role of cyclase-associated protein in regulating actin filament dynamics - more than a monomer-sequestration factor. *J Cell Sci.* 2013; 126:3249–3258. [PubMed: 23908377]
13. Andrianantoandro E, Pollard TD. Mechanism of actin filament turnover by severing and nucleation at different concentrations of ADF/cofilin. *Mol Cell.* 2006; 24:13–23. [PubMed: 17018289]
14. Ono S. Regulation of actin filament dynamics by actin depolymerizing factor/cofilin and actin-interacting protein 1: new blades for twisted filaments. *Biochemistry.* 2003; 42:13363–13370. [PubMed: 14621980]
15. Amberg DC, Basart E, Botstein D. Defining protein interactions with yeast actin in vivo. *Nat Struct Biol.* 1995; 2:28–35. [PubMed: 7719850]
16. Adler HJ, Winnicki RS, Gong TW, Lomax MI. A gene upregulated in the acoustically damaged chick basilar papilla encodes a novel WD40 repeat protein. *Genomics.* 1999; 56:59–69. [PubMed: 10036186]
17. Ren N, Charlton J, Adler PN. The flare gene, which encodes the AIP1 protein of *Drosophila*, functions to regulate F-actin disassembly in pupal epidermal cells. *Genetics.* 2007; 176:2223–2234. [PubMed: 17565945]
18. Ono S. The *Caenorhabditis elegans* unc-78 gene encodes a homologue of actin-interacting protein 1 required for organized assembly of muscle actin filaments. *J Cell Biol.* 2001; 152:1313–1319. [PubMed: 11257131]
19. Waterston RH, Thomson JN, Brenner S. Mutants with altered muscle structure of *Caenorhabditis elegans*. *Dev Biol.* 1980; 77:271–302. [PubMed: 7190524]
20. Ono S, Nomura K, Hitosugi S, Tu DK, Lee JA, Baillie DL, Ono K. The two actin-interacting protein 1 genes have overlapping and essential function for embryonic development in *Caenorhabditis elegans*. *Mol Biol Cell.* 2011; 22:2258–2269. [PubMed: 21551072]
21. Smith TF. Diversity of WD-repeat proteins. *Subcell Biochem.* 2008; 48:20–30. [PubMed: 18925368]
22. Voegtli WC, Madrona AY, Wilson DK. The structure of Aip1p, a WD repeat protein that regulates cofilin-mediated actin depolymerization. *J Biol Chem.* 2003; 278:34373–34379. [PubMed: 12807914]
23. Mohri K, Vorobiev S, Fedorov AA, Almo SC, Ono S. Identification of functional residues on *Caenorhabditis elegans* actin-interacting protein 1 (UNC-78) for disassembly of actin depolymerizing factor/cofilin-bound actin filaments. *J Biol Chem.* 2004; 279:31697–31707. [PubMed: 15150269]
24. Rodal AA, Tetreault JW, Lappalainen P, Drubin DG, Amberg DC. Aip1p interacts with cofilin to disassemble actin filaments. *J Cell Biol.* 1999; 145:1251–1264. [PubMed: 10366597]
25. Okada K, Obinata T, Abe H. XAIP1: a *Xenopus* homologue of yeast actin interacting protein 1 (AIP1), which induces disassembly of actin filaments cooperatively with ADF/cofilin family proteins. *J Cell Sci.* 1999; 112:1553–1565. [PubMed: 10212149]
26. Aizawa H, Katadae M, Maruya M, Sameshima M, Murakami-Murofushi K, Yahara I. Hyperosmotic stress-induced reorganization of actin bundles in *Dictyostelium* cells over-expressing cofilin. *Genes Cells.* 1999; 4:311–324. [PubMed: 10421841]

27. Iida K, Yahara I. Cooperation of two actin-binding proteins, cofilin and Aip1, in *Saccharomyces cerevisiae*. *Genes Cells*. 1999; 4:21–32. [PubMed: 10231390]
28. Okada K, Ravi H, Smith EM, Goode BL. Aip1 and cofilin promote rapid turnover of yeast actin patches and cables: A coordinated mechanism for severing and capping filaments. *Mol Biol Cell*. 2006; 17:2855–2868. [PubMed: 16611742]
29. Clark MG, Amberg DC. Biochemical and genetic analyses provide insight into the structural and mechanistic properties of actin filament disassembly by the Aip1p cofilin complex in *Saccharomyces cerevisiae*. *Genetics*. 2007; 176:1527–1539. [PubMed: 17483419]
30. Clark MG, Teply J, Haarer BK, Viggiano SC, Sept D, Amberg DC. A genetic dissection of Aip1p's interactions leads to a model for Aip1p-cofilin cooperative activities. *Mol Biol Cell*. 2006; 17:1971–1984. [PubMed: 16421248]
31. Mohri K, Ono K, Yu R, Yamashiro S, Ono S. Enhancement of actin-depolymerizing factor/cofilin-dependent actin disassembly by actin-interacting protein 1 is required for organized actin filament assembly in the *Caenorhabditis elegans* body wall muscle. *Mol Biol Cell*. 2006; 17:2190–2199. [PubMed: 16525019]
32. Talman AM, Chong R, Chia J, Svitkina T, Agaisse H. Actin network disassembly powers dissemination of *Listeria monocytogenes*. *J Cell Sci*. 2014; 127:240–249. [PubMed: 24155331]
33. Choi CH, Patel H, Barber DL. Expression of actin-interacting protein 1 suppresses impaired chemotaxis of *Dictyostelium* cells lacking the $\text{Na}^+\text{-H}^+$ exchanger NHE1. *Mol Biol Cell*. 2010; 21:3162–3170. [PubMed: 20668166]
34. Gressin L, Guillotin A, Guerin C, Blanchoin L, Michelot A. Architecture dependence of actin filament network disassembly. *Curr Biol*. 2015; 25:1437–1447. [PubMed: 25913406]
35. Nadkarni AV, Brieher WM. Aip1 destabilizes cofilin-saturated actin filaments by severing and accelerating monomer dissociation from ends. *Curr Biol*. 2014; 24:2749–2757. [PubMed: 25448002]
36. Jansen S, Collins A, Chin SM, Ydenberg CA, Gelles J, Goode BL. Single-molecule imaging of a three-component ordered actin disassembly mechanism. *Nat Commun*. 2015; 6:7202. [PubMed: 25995115]
37. Chen Q, Courtemanche N, Pollard TD. Aip1 promotes actin filament severing by cofilin and regulates constriction of the cytokinetic contractile ring. *J Biol Chem*. 2015; 290:2289–2300. [PubMed: 25451933]
38. Shi M, Xie Y, Zheng Y, Wang J, Su Y, Yang Q, Huang S. *Oryza sativa* actin-interacting protein 1 is required for rice growth by promoting actin turnover. *Plant J*. 2013; 73:747–760. [PubMed: 23134061]
39. Nomura K, Hayakawa K, Tatsumi H, Ono S. Actin-interacting protein 1 promotes disassembly of actin-depolymerizing factor/cofilin-bound actin filaments in a pH-dependent manner. *J Biol Chem*. 2016; 291:5146–5156. [PubMed: 26747606]
40. Shekhar S, Carlier MF. Enhanced depolymerization of actin filaments by ADF/cofilin and monomer funneling by capping protein cooperate to accelerate barbed-end growth. *Curr Biol*. 2017; 27:1990–1998. [PubMed: 28625780]
41. Didry D, Carlier MF, Pantaloni D. Synergy between actin depolymerizing factor/cofilin and profilin in increasing actin filament turnover. *J Biol Chem*. 1998; 273:25602–25611. [PubMed: 9748225]
42. Suarez C, Roland J, Boujemaa-Paterski R, Kang H, McCullough BR, Reymann AC, Guerin C, Martiel JL, De la Cruz EM, Blanchoin L. Cofilin tunes the nucleotide state of actin filaments and severs at bare and decorated segment boundaries. *Curr Biol*. 2011; 21:862–868. [PubMed: 21530260]
43. Pavlov D, Muhrad A, Cooper J, Wear M, Reisler E. Actin filament severing by cofilin. *J Mol Biol*. 2007; 365:1350–1358. [PubMed: 17134718]
44. McCullough BR, Grintsevich EE, Chen CK, Kang H, Hutchison AL, Henn A, Cao W, Suarez C, Martiel JL, Blanchoin L, Reisler E, De La Cruz EM. Cofilin-linked changes in actin filament flexibility promote severing. *Biophys J*. 2011; 101:151–159. [PubMed: 21723825]

45. Hayakawa K, Sakakibara S, Sokabe M, Tatsumi H. Single-molecule imaging and kinetic analysis of cooperative cofilin-actin filament interactions. *Proc Natl Acad Sci USA*. 2014; 111:9810–9815. [PubMed: 24958883]
46. De La Cruz EM. Cofilin binding to muscle and non-muscle actin filaments: isoform-dependent cooperative interactions. *J Mol Biol*. 2005; 346:557–564. [PubMed: 15670604]
47. Ngo KX, Kodera N, Katayama E, Ando T, Uyeda TQ. Cofilin-induced unidirectional cooperative conformational changes in actin filaments revealed by high-speed atomic force microscopy. *eLife*. 2015; 4doi: 10.7554/eLife.04806
48. Wioland H, Guichard B, Senju Y, Myram S, Lappalainen P, Jegou A, Romet-Lemonne G. ADF/cofilin accelerates actin dynamics by severing filaments and promoting their depolymerization at both ends. *Curr Biol*. 2017; 27:1956–1967. [PubMed: 28625781]
49. Elam WA, Kang H, De La Cruz EM. Competitive displacement of cofilin can promote actin filament severing. *Biochem Biophys Res Commun*. 2013; 438:728–731. [PubMed: 23911787]
50. Aggeli D, Kish-Trier E, Lin MC, Haarer B, Cingolani G, Cooper JA, Wilkens S, Amberg DC. Coordination of the filament stabilizing versus destabilizing activities of cofilin through its secondary binding site on actin. *Cytoskeleton (Hoboken)*. 2014; 71:361–379. [PubMed: 24943913]
51. Pope BJ, Gonsior SM, Yeoh S, McGough A, Weeds AG. Uncoupling actin filament fragmentation by cofilin from increased subunit turnover. *J Mol Biol*. 2000; 298:649–661. [PubMed: 10788327]
52. Lappalainen P, Fedorov EV, Fedorov AA, Almo SC, Drubin DG. Essential functions and actin-binding surfaces of yeast cofilin revealed by systematic mutagenesis. *EMBO J*. 1997; 16:5520–5530. [PubMed: 9312011]
53. Ono S, McGough A, Pope BJ, Tolbert VT, Bui A, Pohl J, Benian GM, Gernert KM, Weeds AG. The C-terminal tail of UNC-60B (ADF/cofilin) is critical for maintaining its stable association with F-actin and is implicated in the second actin-binding site. *J Biol Chem*. 2001; 276:5952–5958. [PubMed: 11050090]
54. Chin SM, Jansen S, Goode BL. TIRF microscopy analysis of human Cof1, Cof2, and ADF effects on actin filament severing and turnover. *J Mol Biol*. 2016; 428:1604–1616. [PubMed: 26996939]
55. Ono S, Benian GM. Two *Caenorhabditis elegans* actin depolymerizing factor/cofilin proteins, encoded by the unc-60 gene, differentially regulate actin filament dynamics. *J Biol Chem*. 1998; 273:3778–3783. [PubMed: 9452511]
56. Yamashiro S, Mohri K, Ono S. The two *Caenorhabditis elegans* actin depolymerizing factor/cofilin proteins differently enhance actin filament severing and depolymerization. *Biochemistry*. 2005; 44:14238–14247. [PubMed: 16245940]
57. Shukla VK, Kabra A, Maheshwari D, Yadav R, Jain A, Tripathi S, Ono S, Kumar D, Arora A. Solution structures and dynamics of ADF/cofilins UNC-60A and UNC-60B from *Caenorhabditis elegans*. *Biochem J*. 2015; 465:63–78. [PubMed: 25279657]
58. Mohri K, Ono S. Actin filament disassembling activity of *Caenorhabditis elegans* actin-interacting protein 1 (UNC-78) is dependent on filament binding by a specific ADF/cofilin isoform. *J Cell Sci*. 2003; 116:4107–4118. [PubMed: 12953066]
59. Mizuno K. Signaling mechanisms and functional roles of cofilin phosphorylation and dephosphorylation. *Cell Signal*. 2013; 25:457–469. [PubMed: 23153585]
60. Ohashi K. Roles of cofilin in development and its mechanisms of regulation. *Dev Growth Differ*. 2015; 57:275–290. [PubMed: 25864508]
61. Yonezawa N, Nishida E, Sakai H. pH control of actin polymerization by cofilin. *J Biol Chem*. 1985; 260:14410–14412. [PubMed: 4055781]
62. Bernstein BW, Painter WB, Chen H, Minamide LS, Abe H, Bamburg JR. Intracellular pH modulation of ADF/cofilin proteins. *Cell Motil Cytoskeleton*. 2000; 47:319–336. [PubMed: 11093252]
63. Webb BA, Chimenti M, Jacobson MP, Barber DL. Dysregulated pH: a perfect storm for cancer progression. *Nat Rev Cancer*. 2011; 11:671–677. [PubMed: 21833026]
64. Brieher WM, Kueh HY, Ballif BA, Mitchison TJ. Rapid actin monomer-insensitive depolymerization of *Listeria* actin comet tails by cofilin, coronin, and Aip1. *J Cell Biol*. 2006; 175:315–324. [PubMed: 17060499]

65. Kueh HY, Charras GT, Mitchison TJ, Brieher WM. Actin disassembly by cofilin, coronin, and Aip1 occurs in bursts and is inhibited by barbed-end cappers. *J Cell Biol.* 2008; 182:341–353. [PubMed: 18663144]
66. Li J, Brieher WM, Scimone ML, Kang SJ, Zhu H, Yin H, von Andrian UH, Mitchison T, Yuan J. Caspase-11 regulates cell migration by promoting Aip1-Cofilin-mediated actin depolymerization. *Nat Cell Biol.* 2007; 9:276–286. [PubMed: 17293856]
67. Gandhi M, Achard V, Blanchoin L, Goode BL. Coronin switches roles in actin disassembly depending on the nucleotide state of actin. *Mol Cell.* 2009; 34:364–374. [PubMed: 19450534]
68. Mikati MA, Breitsprecher D, Jansen S, Reisler E, Goode BL. Coronin enhances actin filament severing by recruiting cofilin to filament sides and altering F-actin conformation. *J Mol Biol.* 2015
69. Chan KT, Creed SJ, Bear JE. Unraveling the enigma: progress towards understanding the coronin family of actin regulators. *Trends Cell Biol.* 2011; 21:481–488. [PubMed: 21632254]
70. Lin MC, Galletta BJ, Sept D, Cooper JA. Overlapping and distinct functions for cofilin, coronin and Aip1 in actin dynamics in vivo. *J Cell Sci.* 2010; 123:1329–1342. [PubMed: 20332110]
71. Ishikawa-Ankerhold HC, Gerisch G, Müller-Taubenberger A. Genetic evidence for concerted control of actin dynamics in cytokinesis, endocytic traffic, and cell motility by coronin and Aip1. *Cytoskeleton (Hoboken).* 2010; 67:442–455. [PubMed: 20506401]
72. Okada K, Blanchoin L, Abe H, Chen H, Pollard TD, Bamburg JR. Xenopus actin-interacting protein 1 (XAip1) enhances cofilin fragmentation of filaments by capping filament ends. *J Biol Chem.* 2002; 277:43011–43016. [PubMed: 12055192]
73. Balcer HI, Goodman AL, Rodal AA, Smith E, Kugler J, Heuser JE, Goode BL. Coordinated regulation of actin filament turnover by a high-molecular-weight Srv2/CAP complex, cofilin, profilin, and Aip1. *Curr Biol.* 2003; 13:2159–2169. [PubMed: 14680631]
74. Ono S, Mohri K, Ono K. Microscopic evidence that actin-interacting protein 1 actively disassembles actin-depolymerizing factor/cofilin-bound actin filaments. *J Biol Chem.* 2004; 279:14207–14212. [PubMed: 14742433]
75. Tsuji T, Miyoshi T, Higashida C, Narumiya S, Watanabe N. An order of magnitude faster AIP1-associated actin disruption than nucleation by the Arp2/3 complex in lamellipodia. *PLoS One.* 2009; 4:e4921. [PubMed: 19290054]
76. Berro J, Pollard TD. Synergies between Aip1p and capping protein subunits (Acp1p and Acp2p) in clathrin-mediated endocytosis and cell polarization in fission yeast. *Mol Biol Cell.* 2014; 25:3515–3527. [PubMed: 25143407]
77. Michelot A, Grassart A, Okreglak V, Costanzo M, Boone C, Drubin DG. Actin filament elongation in Arp2/3-derived networks is controlled by three distinct mechanisms. *Dev Cell.* 2013; 24:182–195. [PubMed: 23333351]
78. Ono K, Ono S. Actin-ADF/Cofilin Rod Formation in *Caenorhabditis elegans* Muscle Requires a Putative F-Actin Binding Site of ADF/Cofilin at the C-Terminus. *Cell Motil Cytoskeleton.* 2009; 66:398–408. [PubMed: 19459188]
79. Ishikawa-Ankerhold HC, Daszkiewicz W, Schleicher M, Müller-Taubenberger A. Actin-interacting protein 1 contributes to intranuclear rod assembly in *Dictyostelium discoideum*. *Sci Rep.* 2017; 7:40310. [PubMed: 28074884]
80. Bernstein BW, Chen H, Boyle JA, Bamburg JR. Formation of actin-ADF/cofilin rods transiently retards decline of mitochondrial potential and ATP in stressed neurons. *Am J Physiol Cell Physiol.* 2006; 291:C828–C839. [PubMed: 16738008]
81. Kile BT, Panopoulos AD, Stirzaker RA, Hacking DF, Tahtamouni LH, Willson TA, Mielke LA, Henley KJ, Zhang JG, Wicks IP, Stevenson WS, Nurden P, Watowich SS, Justice MJ. Mutations in the cofilin partner Aip1/Wdr1 cause autoinflammatory disease and macrothrombocytopenia. *Blood.* 2007; 110:2371–2380. [PubMed: 17515402]
82. Xiao Y, Ma H, Wan P, Qin D, Wang X, Zhang X, Xiang Y, Liu W, Chen J, Yi Z, Li L. Trp-Asp (WD) repeat domain 1 is essential for mouse peri-implantation development and regulates cofilin phosphorylation. *J Biol Chem.* 2017; 292:1438–1448. [PubMed: 27994054]
83. Ketelaar T, Allwood EG, Anthony R, Voigt B, Menzel D, Hussey PJ. The actin-interacting protein AIP1 is essential for actin organization and plant development. *Curr Biol.* 2004; 14:145–149. [PubMed: 14738737]

84. Kuhns DB, Fink DL, Choi U, Sweeney C, Lau K, Priel DL, Riva D, Mendez L, Uzel G, Freeman AF, Olivier KN, Anderson VL, Currens R, Mackley V, Kang A, Al-Adeli M, Mace E, Orange JS, Kang E, Lockett SJ, Chen, Steinbach PJ, Hsu AP, Zarembek KA, Malech HL, Gallin JI, Holland SM. Cytoskeletal abnormalities and neutrophil dysfunction in WDR1 deficiency. *Blood*. 2016; 128:2135–2143. [PubMed: 27557945]
85. Standing AS, Malinova D, Hong Y, Record J, Moulding D, Blundell MP, Nowak K, Jones H, Omoyinmi E, Gilmour KC, Medlar A, Stanescu H, Kleta R, Anderson G, Nanthapisal S, Gomes SM, Klein N, Eleftheriou D, Thrasher AJ, Brogan PA. Autoinflammatory periodic fever, immunodeficiency, and thrombocytopenia (PFIT) caused by mutation in actin-regulatory gene WDR1. *J Exp Med*. 2017; 214:59–71. [PubMed: 27994071]
86. Oh SH, Adler HJ, Raphael Y, Lomax MI. WDR1 colocalizes with ADF and actin in the normal and noise-damaged chick cochlea. *J Comp Neurol*. 2002; 448:399–409. [PubMed: 12115702]
87. Chu D, Pan H, Wan P, Wu J, Luo J, Zhu H, Chen J. AIP1 acts with cofilin to control actin dynamics during epithelial morphogenesis. *Development*. 2012; 139:3561–3571. [PubMed: 22899846]
88. Ono K, Ono S. Two actin-interacting protein 1 isoforms function redundantly in the somatic gonad and are essential for reproduction in *Caenorhabditis elegans*. *Cytoskeleton*. 2014; 71:36–45. [PubMed: 24130131]
89. Luxenburg C, Heller E, Pasolli HA, Chai S, Nikolova M, Stokes N, Fuchs E. Wdr1-mediated cell shape dynamics and cortical tension are essential for epidermal planar cell polarity. *Nat Cell Biol*. 2015; 17:592–604. [PubMed: 25915128]
90. Lechuga S, Baranwal S, Ivanov AI. Actin-interacting protein 1 controls assembly and permeability of intestinal epithelial apical junctions. *Am J Physiol Gastrointest Liver Physiol*. 2015; 308:G745–756. [PubMed: 25792565]
91. Schnorrer F, Schonbauer C, Langer CC, Dietzl G, Novatchkova M, Schernhuber K, Fellner M, Azaryan A, Radolf M, Stark A, Keleman K, Dickson BJ. Systematic genetic analysis of muscle morphogenesis and function in *Drosophila*. *Nature*. 2010; 464:287–291. [PubMed: 20220848]
92. Yuan B, Wan P, Chu D, Nie J, Cao Y, Luo W, Lu S, Chen J, Yang Z. A cardiomyocyte-specific Wdr1 knockout demonstrates essential functional roles for actin disassembly during myocardial growth and maintenance in mice. *Am J Pathol*. 2014; 184:1967–1980. [PubMed: 24840128]
93. Ono S. Dynamic regulation of sarcomeric actin filaments in striated muscle. *Cytoskeleton (Hoboken)*. 2010; 67:677–692. [PubMed: 20737540]
94. Yamashiro S, Cox EA, Baillie DL, Hardin JD, Ono S. Sarcomeric actin organization is synergistically promoted by tropomodulin, ADF/cofilin, AIP1 and profilin in *C. elegans*. *J Cell Sci*. 2008; 121:3867–3877. [PubMed: 18984629]
95. Salanova M, Gelfi C, Moriggi M, Vasso M, Vigano A, Minafra L, Bonifacio G, Schiffl G, Gutschmann M, Felsenberg D, Cerretelli P, Blottnner D. Disuse deterioration of human skeletal muscle challenged by resistive exercise superimposed with vibration: evidence from structural and proteomic analysis. *FASEB J*. 2014; 28:4748–4763. [PubMed: 25122557]
96. Kim ML, Chae JJ, Park YH, De Nardo D, Stirzaker RA, Ko HJ, Tye H, Cengia L, DiRago L, Metcalf D, Roberts AW, Kastner DL, Lew AM, Lyras D, Kile BT, Croker BA, Masters SL. Aberrant actin depolymerization triggers the pyrin inflammasome and autoinflammatory disease that is dependent on IL-18, not IL-1 β . *J Exp Med*. 2015; 212:927–938. [PubMed: 26008898]
97. Montenont E, Echagarruga C, Allen N, Araldi E, Suarez Y, Berger JS. Platelet WDR1 suppresses platelet activity and is associated with cardiovascular disease. *Blood*. 2016; 128:2033–2042. [PubMed: 27609643]
98. Dasgupta SK, Le A, Da Q, Cruz M, Rumbaut RE, Thiagarajan P. Wdr1-dependent actin reorganization in platelet activation. *PLoS One*. 2016; 11:e0162897. [PubMed: 27627652]
99. Southwick FS. The lazy leukocyte syndrome revisited. *Blood*. 2016; 128:2112–2113. [PubMed: 27789436]
100. Wang W, Eddy R, Condeelis J. The cofilin pathway in breast cancer invasion and metastasis. *Nat Rev Cancer*. 2007; 7:429–440. [PubMed: 17522712]
101. Bravo-Cordero JJ, Magalhaes MA, Eddy RJ, Hodgson L, Condeelis J. Functions of cofilin in cell locomotion and invasion. *Nat Rev Mol Cell Biol*. 2013; 14:405–415. [PubMed: 23778968]

102. Lee JH, Kim JE, Kim BG, Han HH, Kang S, Cho NH. STAT3-induced WDR1 overexpression promotes breast cancer cell migration. *Cell Signal*. 2016; 28:1753–1760. [PubMed: 27521604]
103. Park K, Lee JH, Han HH, Mun SG, Kang S, Cha YJ, Koo JS, Kim MJ, Lee HS, Moon J, Cho NH. Nodal metastasis signatures in breast cancer. *Pathol Res Pract*. 2017; 213:680–687. [PubMed: 28476377]
104. Kang S, Kim MJ, An H, Kim BG, Choi YP, Kang KS, Gao MQ, Park H, Na HJ, Kim HK, Yun HR, Kim DS, Cho NH. Proteomic molecular portrait of interface zone in breast cancer. *J Proteome Res*. 2010; 9:5638–5645. [PubMed: 20857901]
105. Kim DH, Bae J, Lee JW, Kim SY, Kim YH, Bae JY, Yi JK, Yu MH, Noh DY, Lee C. Proteomic analysis of breast cancer tissue reveals upregulation of actin-remodeling proteins and its relevance to cancer invasiveness. *Proteomics Clin Appl*. 2009; 3:30–40. [PubMed: 21136934]
106. Xu H, Chen Y, Tan C, Xu T, Yan Y, Qin R, Huang Q, Lu C, Liang C, Lu Y, Wang H, Chen J. High expression of WDR1 in primary glioblastoma is associated with poor prognosis. *Am J Transl Res*. 2016; 8:1253–1264. [PubMed: 27158412]
107. Huang YT, Zhang Y, Wu Z, Michaud DS. Genotype-based gene signature of glioma risk. *Neuro Oncol*. 2017; 19:940–950. [PubMed: 28339748]
108. Haslene-Hox H, Oveland E, Woie K, Salvesen HB, Wiig H, Tenstad O. Increased WD-repeat containing protein 1 in interstitial fluid from ovarian carcinomas shown by comparative proteomic analysis of malignant and healthy gynecological tissue. *Biochim Biophys Acta*. 2013; 1834:2347–2359. [PubMed: 23707566]
109. Izawa S, Okamura T, Matsuzawa K, Ohkura T, Ohkura H, Ishiguro K, Noh JY, Kamijo K, Yoshida A, Shigemasa C, Kato M, Yamamoto K, Taniguchi S. Autoantibody against WD repeat domain 1 is a novel serological biomarker for screening of thyroid neoplasia. *Clin Endocrinol (Oxf)*. 2013; 79:35–42. [PubMed: 23215816]
110. Xiang Y, Liao XH, Yao A, Qin H, Fan LJ, Li JP, Hu P, Li H, Guo W, Li JY, Gu CJ, Bao LY, Zhang TC. MRTF-A-miR-206-WDR1 form feedback loop to regulate breast cancer cell migration. *Exp Cell Res*. 2017:394–404.
111. Kato A, Kurita S, Hayashi A, Kaji N, Ohashi K, Mizuno K. Critical roles of actin-interacting protein 1 in cytokinesis and chemotactic migration of mammalian cells. *Biochem J*. 2008; 414:261–270. [PubMed: 18494608]
112. Konzok A, Weber I, Simmeth E, Hacker U, Maniak M, Müller-Taubenberger A. DAip1, a Dictyostelium homologue of the yeast actin-interacting protein 1, is involved in endocytosis, cytokinesis, and motility. *J Cell Biol*. 1999; 146:453–464. [PubMed: 10427097]
113. Rogers SL, Wiedemann U, Stuurman N, Vale RD. Molecular requirements for actin-based lamella formation in Drosophila S2 cells. *J Cell Biol*. 2003; 162:1079–1088. [PubMed: 12975351]
114. Poukkula M, Hakala M, Penttimikko N, Sweeney MO, Jansen S, Mattila J, Hietakangas V, Goode BL, Lappalainen P. GMF promotes leading-edge dynamics and collective cell migration in vivo. *Curr Biol*. 2014; 24:2533–2540. [PubMed: 25308079]
115. Fujibuchi T, Abe Y, Takeuchi T, Imai Y, Kamei Y, Murase R, Ueda N, Shigemoto K, Yamamoto H, Kito K. AIP1/WDR1 supports mitotic cell rounding. *Biochem Biophys Res Commun*. 2005; 327:268–275. [PubMed: 15629458]
116. Okreglak V, Drubin DG. Loss of Aip1 reveals a role in maintaining the actin monomer pool and an in vivo oligomer assembly pathway. *J Cell Biol*. 2010; 188:769–777. [PubMed: 20231387]
117. Virtanen JA, Vartiainen MK. Diverse functions for different forms of nuclear actin. *Curr Opin Cell Biol*. 2017; 46:33–38. [PubMed: 28092729]
118. Allwood EG, Anthony RG, Smertenko AP, Reichelt S, Drobak BK, Doonan JH, Weeds AG, Hussey PJ. Regulation of the pollen-specific actin-depolymerizing factor LIADF1. *Plant Cell*. 2002; 14:2915–2927. [PubMed: 12417710]
119. Ketelaar T, Allwood EG, Hussey PJ. Actin organization and root hair development are disrupted by ethanol-induced overexpression of Arabidopsis actin interacting protein 1 (AIP1). *New Phytol*. 2007; 174:57–62. [PubMed: 17335497]
120. Kiefer CS, Claes AR, Nzayisenga JC, Pietra S, Stanislas T, Huser A, Ikeda Y, Grebe M. Arabidopsis AIP1-2 restricted by WER-mediated patterning modulates planar polarity. *Development*. 2015; 142:151–161. [PubMed: 25428588]

121. Lovy-Wheeler A, Kunkel JG, Allwood EG, Hussey PJ, Hepler PK. Oscillatory increases in alkalinity anticipate growth and may regulate actin dynamics in pollen tubes of lily. *Plant Cell*. 2006; 18:2182–2193. [PubMed: 16920777]
122. Augustine RC, Pattavina KA, Tuzel E, Vidali L, Bezanilla M. Actin interacting protein1 and actin depolymerizing factor drive rapid actin dynamics in *Physcomitrella patens*. *Plant Cell*. 2011; 23:3696–3710. [PubMed: 22003077]
123. Matsumoto S, Ogawa M, Kasakura T, Shimada Y, Mitsui M, Maruya M, Isohata M, Yahara I, Murakami-Murofushi K. A novel 66-kDa stress protein, p66, associated with the process of cyst formation of *Physarum polycephalum* is a *Physarum* homologue of a yeast actin-interacting protein, AIP1. *J Biochem (Tokyo)*. 1998; 124:326–331. [PubMed: 9685722]

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Highlights

- AIP1 promotes disassembly of ADF/cofilin-decorated actin filaments.
- AIP1 contains 14 WD-repeats forming two β -propeller domains in a single molecule.
- AIP1 deficiency often causes lethality that is primarily due to abnormalities in the actin cytoskeleton.

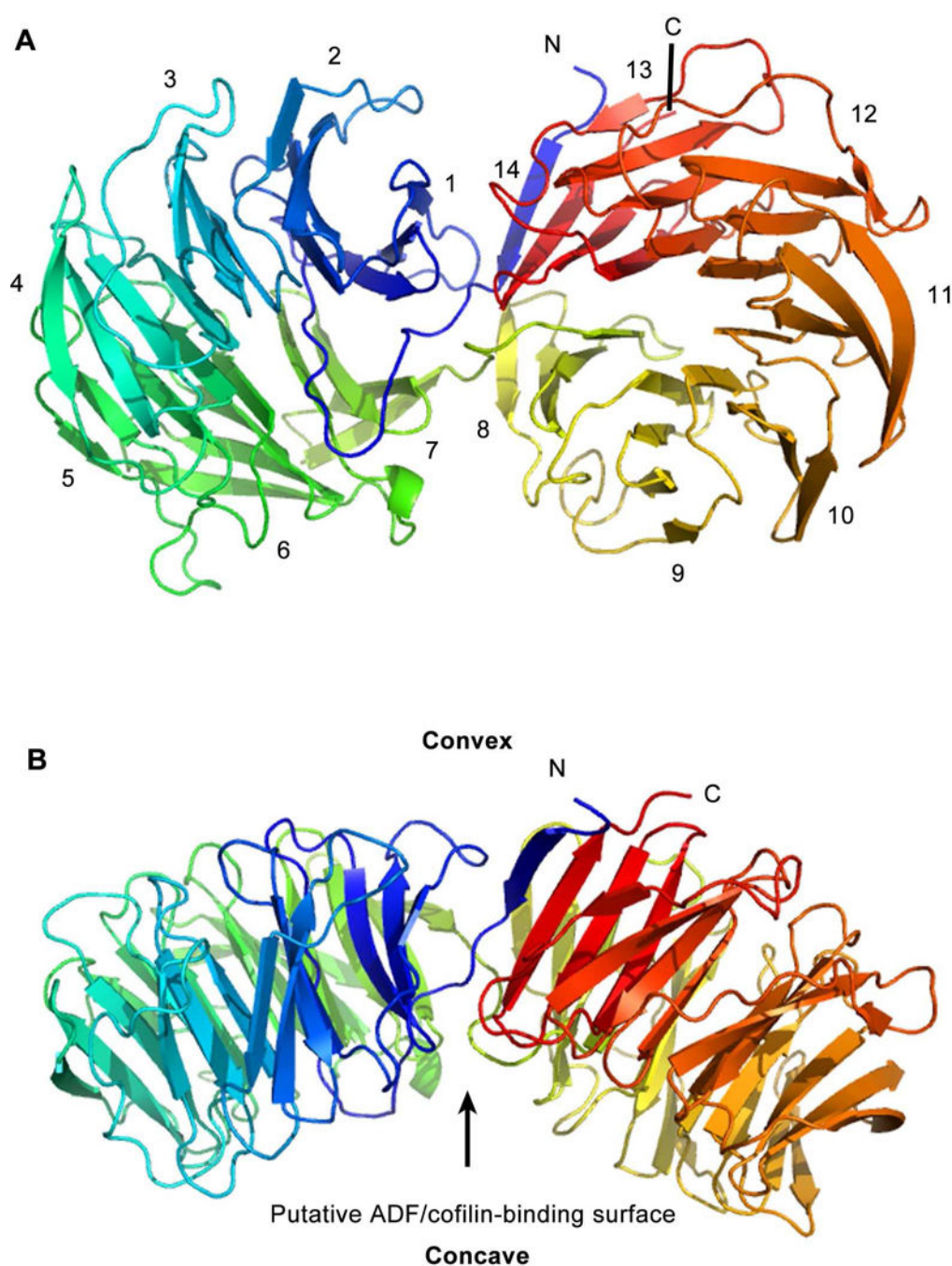


Figure 1. Structure of AIP1. The crystal structure of *C. elegans* UNC-78 (Protein Data Bank accession code 1PEV) is shown from the concave surface (A) and the side of β -propellers (B). Numbers of WD-repeat blades are indicated in A. N- and C-termini are indicated as N and C, respectively. The molecular graphics are generated using PyMol (Schrödinger).

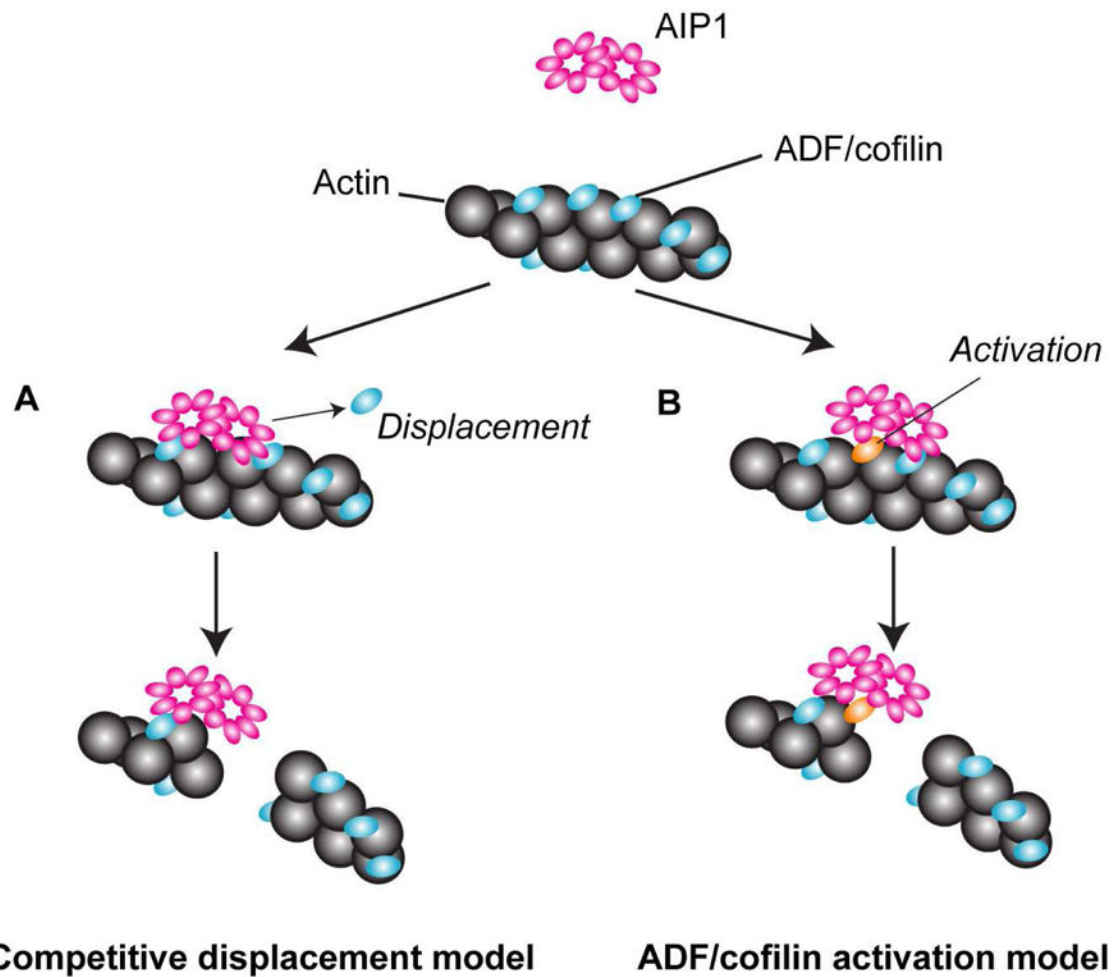


Figure 2.

Hypothetical mechanisms of actin filament severing by AIP1. (A) In the competitive displacement model, AIP1 competitively displaces ADF/cofilin and generates a boundary between ADF/cofilin-bound and unbound region, which is susceptible for fragmentation. (B) In the ADF/cofilin activation model, AIP1 induces a conformational change of ADF/cofilin (shown in orange) to enhance its filament severing activity.

Table 1

Representative AIP1 genes and proteins in eukaryotes

Class	Organism	Species	Gene	Protein	Reference
Vertebrates	Mammals, birds	<i>Homo sapiens, Mus musculus, Gallus gallus</i>	<i>WDR1</i>	WDR1/AIP1	[16]
Vertebrates	Frog	<i>Xenopus laevis</i>	– –	XAIP1a ¹ XAIP1b ¹	[25]
Arthropods	Insect (fruit fly)	<i>Drosophila melanogaster</i>	<i>flare</i>	Flare (Flt)	[17]
Nematodes	Roundworm	<i>Caenorhabditis elegans</i>	<i>unc-78</i> <i>aip1-1</i>	UNC-78 AIP1-1	[18] [20]
Fungi	Budding yeast	<i>Saccharomyces cerevisiae</i>	<i>aip1</i>	Aip1p	[15]
Fungi	Fission yeast	<i>Schizosaccharomyces pombe</i>	<i>aip1</i>	Aip1p	[76]
Amoebozoa	Slime mold	<i>Dictyostelium discoideum</i>	<i>DAIP1</i>	DAIP1	[112]
Amoebozoa	Slime mold	<i>Physarum polycephalum</i>	–	p66	[123]
Plant	Rockcress	<i>Arabidopsis thaliana</i>	<i>AtAIP1-1</i> <i>AtAIP1-2</i>	AtAIP1-1 AtAIP1-2	[118]
Plant	Rice	<i>Oryza sativa</i>	<i>OsAIP1</i>	OsAIP1	[38]
Plant	Moss	<i>Physcomitrella patens</i>	<i>AIP1</i>	AIP1	[122]

¹The original study by Okada et al (1999) [25] reports the sequence of XAIP1a as XAIP1 (GenBank accession number: AF124140). The sequence of a second AIP1, XAIP1b, is only reported in the database as WD repeat-containing protein 1-B (NCBI Reference Sequence: NP_001083198).